

Worksheet - Lab 2

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Objective

The objective of this experiment is to design and train a Convolutional Neural Network (CNN) on the CIFAR-10 dataset and analyze its computational and training behavior. The experiment focuses on:

- Implementing a custom CNN model
 - Creating a custom data loader for CIFAR-10
 - Estimating the model's computational cost using FLOPs
 - Training the model for 25–30 epochs
 - Visualizing gradient flow and weight update flow
 - Logging and analyzing all results using Weights & Biases (WandB)
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Dataset Description

The CIFAR-10 dataset consists of 60,000 color images of size $32 \times 32 \times 3$, distributed across 10 classes (Airplane, Automobile, Bird, Cat, Deer, Dog, Frog, Horse, Ship, Truck)

The dataset is split into 50,000 training images and 10,000 test images

Custom DataLoader

A custom PyTorch Dataset class was implemented to load CIFAR-10 data.

Key features of the custom DataLoader include:

- Explicit implementation of `__len__()` and `__getitem__()`
 - Data augmentation using random horizontal flipping and random cropping with padding
 - Normalization using CIFAR-10 channel-wise mean and standard deviation
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Model Architecture

A custom Convolutional Neural Network (CNN) was designed instead of using a pretrained architecture to allow better interpretability of gradient and weight flow.

Architecture Overview:

- 3 Convolutional layers with ReLU activation
 - MaxPooling after each convolution block
 - 2 Fully Connected layers
 - Final output layer with 10 neurons (one per class)
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FLOPs Analysis

The number of Floating Point Operations (FLOPs) for the model was computed using a model complexity analysis tool.

Layer	Parameters	MACs	FLOPs	% of Total MACs
conv1 (Conv2d)	896	917.5 K	1.835 M	8.28%
conv2 (Conv2d)	18,500	4.73 M	9.46 M	42.74%
conv3 (Conv2d)	73,860	4.73 M	9.46 M	42.67%
pool (MaxPool2d)	0	57.34 K	114.68 K	0.52%
fc1 (Linear)	524,540	524.54 K	1.049 M	4.74%
fc2 (Linear)	2,570	2.57 K	5.14 K	0.02%
Total	620,360	11.08 M	22.16 M	100%

Observations

- FLOPs increase significantly in convolutional layers compared to fully connected layers.
 - The majority of computation occurs in deeper convolution layers due to increased channel depth.
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Training Setup

Training Configuration

- Optimizer: Adam
 - Loss Function: Cross-Entropy Loss
 - Batch Size: 128
 - Learning Rate: 0.001
 - Epochs: 30
 - Hardware: GPU
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Gradient Flow Visualization

Gradient flow analysis was performed by logging the L2 norm of gradients for each trainable layer at the end of every step/epoch using WandB.



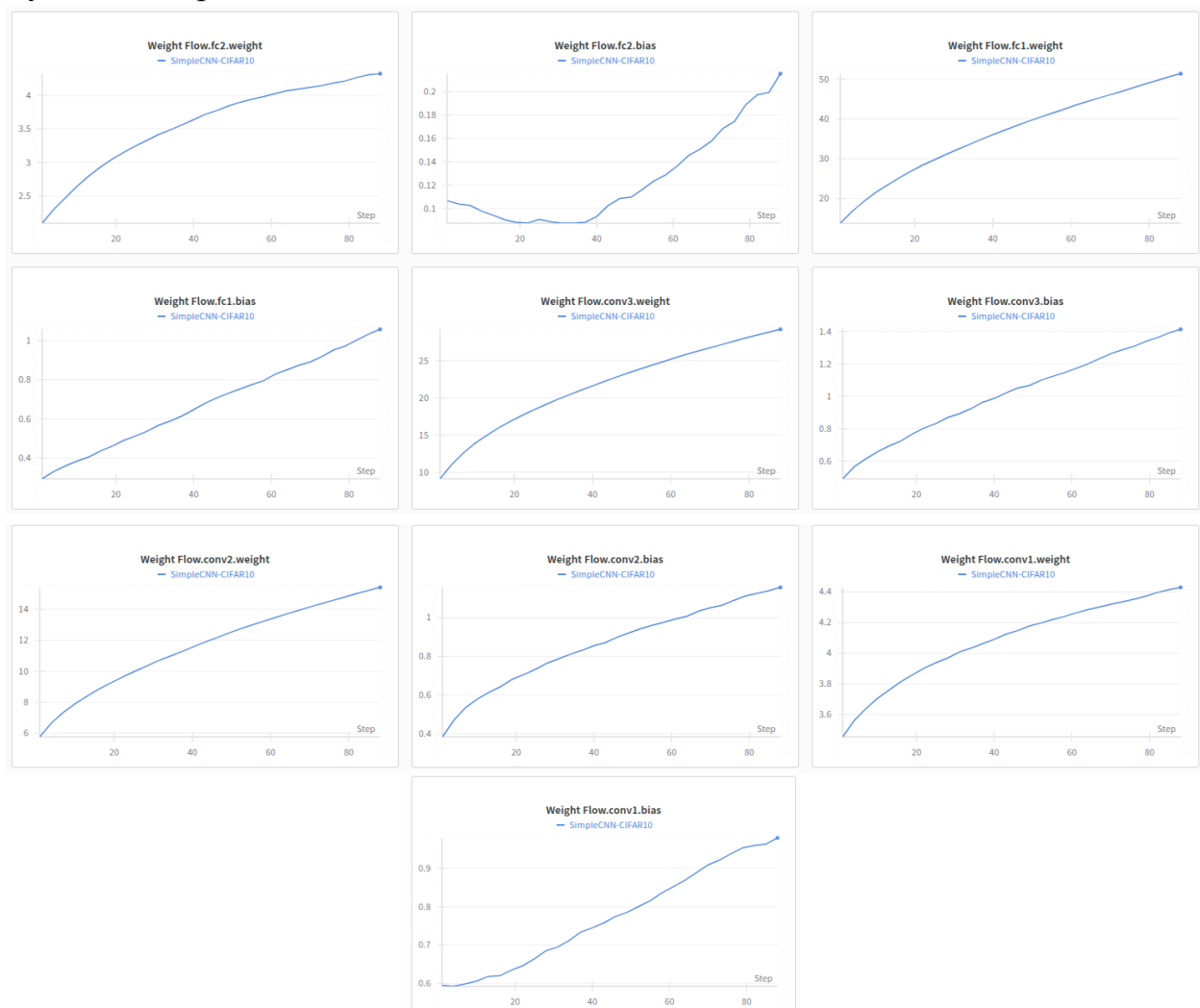
Observations:

- Gradients show noticeable fluctuations across steps rather than being perfectly stable.
- No extreme vanishing or exploding gradients were observed, but short-term oscillations are present.

Although the gradients are not perfectly smooth, the network is still converging effectively, as the weight magnitudes grow gradually and training accuracy improves. The fluctuations are expected in smaller CNNs trained with Adam on CIFAR-10.

Weight Update Flow Analysis

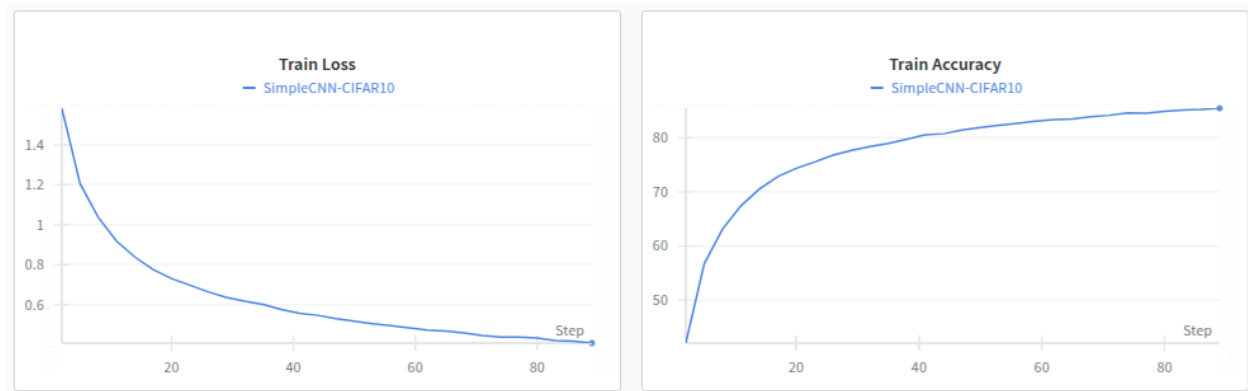
Weight update flow was visualized by logging the **L2 norm of weights** for each layer across epochs.



Observations

- Weight norms increased gradually during early epochs and stabilized later.
 - No sudden spikes or oscillations were observed.
 - Stable weight evolution suggests smooth convergence and effective learning.
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Results



Observations:

- Training loss decreased consistently across epochs.
 - Training accuracy improved steadily, indicating effective learning.
 - Train Accuracy - 85.45%
 - Test Accuracy - 81.07%
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Submission Links

- **GitHub Repository (Branch):**
https://github.com/arpitadeshmukh/MLOps-Arpita_Abhijit_Deshmukh-B23CM1007/tree/Arpita_Abhijit_Deshmukh_B23CM1007_lab2_worksheet-folder
- **Weights & Biases Project:**
<https://wandb.ai/b23cm1007-indian-institute-of-technology-jodhpur/cifar10-cnn-lab2?nw=nwuserb23cm1007>